

MODULE 1: HCI FOUNDATIONS
WEEK 1

Learning Outcomes:

After completing this course you are expected to explore:

1. The roots of human computer interaction as a discipline
2. The various trends which have marked its development and future challenges for research

A. Engage

Quotation Station:

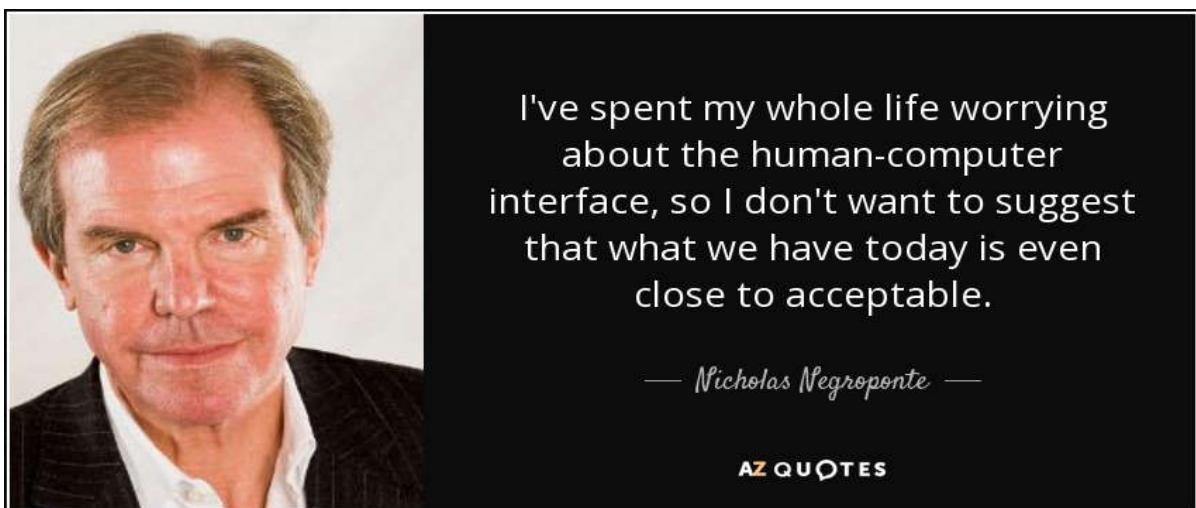


Figure 1.0: Nicholas Negroponte is a Greek American architect. He is the founder and chairman Emeritus of Massachusetts Institute of Technology's Media Lab, and also founded the One Laptop per Child Association. Negroponte is the author of the 1995 bestseller *Being Digital* translated into more than forty languages.

B. Explore

Video Title: What is HCI?

YouTube Link: <https://youtu.be/KtwwustmEDl>

Module Video Filename: Week1_What is HCI?

Additional Reading Resources: Week1_hci_intro.pdf

C. Explain

In this module, we need to provide a general context for both by giving a brief historical view and interest in human-computer interaction (HCI) research. We then integrate various HCI issues into an overarching framework that can encompass broad HCI concerns from multiple disciplines, together with a variety of additional ideas, evidence, and insights. Topics in this module include different disciplines' perspectives on HCI.

D. Elaborate

Introduction

From 'foundations' to 'new paradigms' is a wide canvas and this paper attempts to paint a picture of human–computer interaction from its earliest roots to future challenges. It is also iconic in that HCI as an academic discipline has always been positioned, sometimes uneasily, sometimes creatively, in the tension between solid intellectual rigor and the excitement in new technology. Stefano Levialdi, in who this special issue is in honor, had a rich appreciation of both and so I hope this paper is one that he would have enjoyed as well as offering an overview of the field as it was, as it is and as it could be.

Foundations

Human–computer interaction, like any vocational discipline, sits upon three broad foundations.

Principles

First, and most obviously are the intellectual theories, models and empirical investigations that underlie the field. Give HCI's cross-disciplinary nature, some of these come from a number of related disciplines and some as core HCI knowledge.

Practice

Second, HCI is a field that, inter alia, seeks to offer practical guidance to practitioners in interaction design, usability, UX, or whatever becomes the next key term. However, also it is a discipline that has always sought to learn from the practical design and innovations that surround it.

People

Finally, there are the visionaries who inspire the field and perhaps most importantly the HCI community itself: researchers, educators and practitioners.

Historic Roots

HCI developed as a discipline and a community in the early 1980s, triggered largely by the PC revolution and the mass use of office computers. It was in the early 1980s when the major HCI conferences began Interact, CHI, British HCI and Vienna HCI; all but the last still active today. Core concepts were also formulated in those days including the notion of direct manipulation and user centered design [Sc83, ND86].

The graphical user interface and desktop metaphor, embodied in the early Apple Mac, were the result of work at Xerox PARC throughout the 1970s, mostly based around graphical programming environments such as Smalltalk and Inter Lisp, and leading to the design of the Xerox Star office computer [SK82, JR89], a conceptual breakthrough albeit a commercial failure.

The very first true HCI paper dates back into the late 1950s, with Shackel's 'Ergonomics for a Computer' [Sh59]. While Sutherland and Englebert were early examples of the vision/innovation side of HCI, Shackel's first HCI paper came more from a practical design perspective, the redesign of the control panel of EMIAC II, an early analogue computer.

Sutherland's Sketchpad [Su63], the first graphical user interface dates back to the early 1960s and in the same period, Englebert's 'research centre for augmenting human intellect' was responsible for the invention of the mouse, as well as early versions of electronic conferencing [EE68].

Theory and Contributing Disciplines

In the earliest days the main disciplines involved in HCI were computer science, psychology and ergonomics, as reflected in Shackel's early paper. These disciplines were soon joined by social science, or, to be more precise, the ethnographic and anthropological side of sociology.

The input from ergonomics was initially in terms of physical ergonomics, sitting at a computer terminal, pressing keys; however, this more physical side of HCI declined rapidly as computers became commoditized as opposed to being in special settings and issues of physical ergonomics were relegated to health and safety concerns.

- To some extent this followed from the natural development of the area, once computers were mass-produced, practitioners had little control of the physical system unless they worked for major manufacturers.
- The fields of information science and HCI have been largely parallel rather than interconnected except in the Nordic HCI tradition.
- There are signs of resurfacing interest in more organizational or community focused views of human activity.
- The most marked in the related web science community where methods such as social network analysis are clearly very important, enabled by the vast quantities of data obtained from web-based systems.

The web science 'butterfly', which includes all the disciplines that web science draws on. This was remarkably similar to those that connect with HCI differing mostly in the 'heat map' of those most active or relevant (see figs 1 and 2).

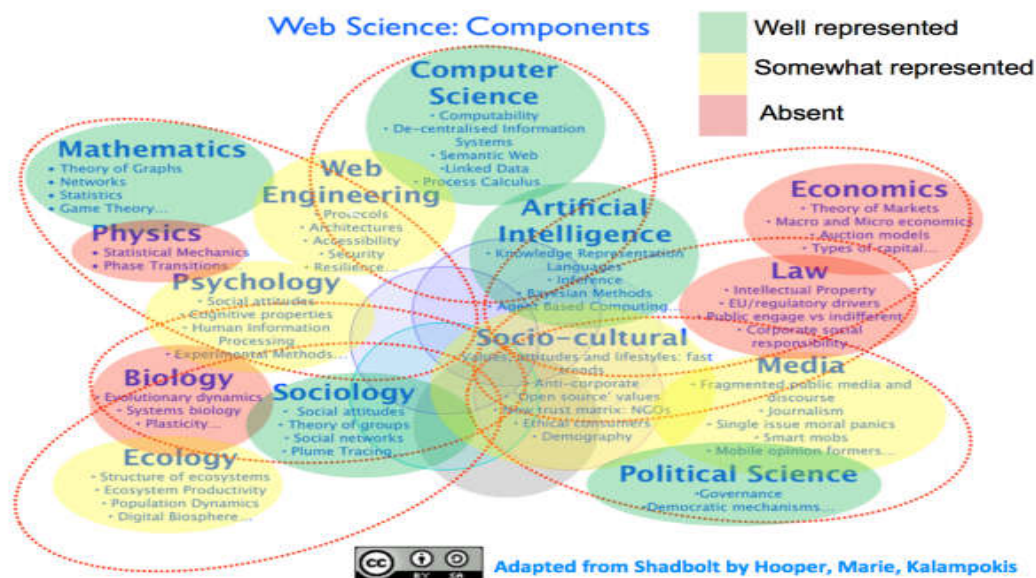


Figure 1.1 Web Science ‘heat map’, showing discipline presence [HM12]



Figure 1.2 Heat map for HCI (from [HD13])

Borrowings from other fields have been very powerful to enable both theoretical and practical interventions. For example, Fitts' Law [Fi54] has created its own small sub-community, human–human conversation analysis has been used to design human–computer dialogues [FL90], and Csikszentmihályi's concept of Flow [Cz90] has proved influential in user experience design. HCI can draw on the methods and knowledge of related fields directly, there are limits to this for two reasons:

Different concerns – The questions we ask in HCI are typically more applied and hence more complex in terms of interrelations than 'base' disciplines, notably psychology. For example, early studies of on-screen reading comprehension, or more recent comparisons of reading comprehension when holding a screen vs with hands on the table [BJ11]; while in many ways

these could be seen as standard perceptual and cognitive psychology, the reasons for studying both were practical and unlikely to have arisen purely from a psychological interest

Integrating knowledge – Some concerns really cut across disciplines requiring theoretical or practical knowledge from multiple areas of study. For example work on how design affects behavior [PV15], requires both behavioral psychology and interaction design.

- There is copious empirical work of this kind; it is harder to find truly integrative HCI theory. There were early descriptive accounts, notably Norman's seven stages of action [No86, No88], and more predictive modeling approaches such as Card, Moran and Newells' 'Model Human Processor' [CM86] and Barnard's 'Interacting Cognitive Subsystems' [Ba85], but, while the former is still influential, there is no clear path of deepening theory of interaction.
- Liu et al. [LG14] used co-word analysis techniques to investigate the development of themes and bodies of knowledge in HCI.
- This showed positive things, notably the level of cross cutting integration across the field; that is there are no disconnected camps.
- It also gave evidence to known effects such as the way that new technologies seem to buffet the discipline starting new themes, which rise for a period before trailing off.
- It also exposed a concerning dearth of integrating bodies of knowledge: "As it stands, the only tradition in HCI is that of having no tradition in terms of research topics. ... when a new technology comes along it seems that researchers start from scratch leading to relatively isolated research themes" [LG14].

E. Evaluate

ASSESSMENT:

Instructions: You may write your answer on the Answer Sheet (AS) provided in this module. (1-point each).

- 1. What year HCI developed as a discipline and a community?
- 2. The first graphical user interface.
- 3. It is used co-word analysis techniques to investigate the development of themes and bodies of knowledge in HCI.
- 4. The redesign of the control panel of EMIAC II, an early analogue computer.
- 5. What year that the early Apple Mac was the result of work at Xerox PARC?
- 6-7. The three broad foundations.
- 9-10. The two reasons limiting the HCI methods and knowledge of related fields.

References

1. https://lecturenotes.in/notes/14593-note-for-human-computer-interaction-hci-by-intu-heroes?reading=true&fbclid=IwAR2mk_8stUtT1n9Cic6fn1wiKO6Y9mjgs9ry-j2Te01hL00c2qfPvLPkmKg

2. <https://www.semanticscholar.org/paper/Human-computer-interaction%2C-foundations-and-new-Dix/2a68487ada287e6ce8111179451fcab4df4b87e>

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